

Animesh Srivastava

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Education

University of Southern California | Los Angeles, USA
Master of Science in Computer Science (GPA: 3.7/4)

August 2023 – May 2025

MS Ramaiah University of Applied Sciences | Bangalore, India
Bachelor of Technology in Computer Science and Engineering

August 2017 – June 2021

Work Experience

Software Engineer, Easley Dunn Productions | Los Angeles, USA

June 2025 – Present

- Built VR surgical training prototype in Unity(C#) with JSON-based session replay & diegetic UI, improving user interaction by 20%.

Full Stack Software Engineer(Founding Engineer), Redvest | Los Angeles, USA

May 2024 – October 2024

- Architected multithreaded **Python** ingestion pipelines for real-time financial analytics, reducing end-to-end data latency by 40%.
- Designed and deployed containerized backend services using **Node.js, TypeScript and Python** on **AWS EKS**, implementing scalable KPI monitoring systems that eliminated 60% of manual reconciliation work.
- Led core feature development across backend and API layers, improving efficiency and boosting user engagement by 20%.
- Developed real-time analytics dashboards using **React** and **TypeScript** to visualize financial and KPI metrics, enabling data-driven product decisions and supporting product presentation at **TechCrunch Disrupt 2024**.

Software Engineer, American Express(via TCS) | Bangalore, India

January 2021 – July 2023

Worked on two projects, **IDL (Indian Data Localization)** and **Travel and Lifestyle Services**

- Engineered distributed ETL pipelines using **Python, PySpark, PL/SQL, and Hive**, implementing automated cross-system validation and reconciliation checks to achieve **100% daily consistency** across **OFSAA, Oracle, and MongoDB**.
- Automated RBI data localization compliance by migrating 30+ validation rules in **ETL pipelines**, reducing regulatory risk by 50%.
- Optimized high-volume **PL/SQL** procedures and ETL jobs by refactoring queries, indexing critical tables, and introducing data quality checkpoints, reducing recurring production failures by 20%.
- Built internal **REST APIs** and multithreaded **Java** automation tools supporting **50K+ daily transactions**, automating report comparisons and saving **20+ engineering hours/week**.

Projects

Animatrix AI Agent - Product Review Aggregator

- Built a **distributed AI product** intelligence system that aggregates multi-source product reviews and pricing data, performs **LLM-based** sentiment and feature extraction, and generates structured product insights in near real time.
- Designed a **Retrieval-Augmented Generation (RAG)** pipeline using **Qdrant** for vector search over embedded review corpora, enabling context-aware summarization and feature comparison grounded in retrieved evidence.
- Implemented **LLM-driven AI agent** task orchestration with **Celery** and **Redis** to coordinate multi-step workflows (scraping, embedding generation, retrieval, LLM reasoning, and price normalization), ensuring scalable and fault-tolerant AI processing.

US Stock Market Portfolio

- Designed and implemented **RESTful** backend services in **Node.js, TypeScript** and **Express** for real-time stock data ingestion, portfolio tracking, and historical price analytics, optimizing MongoDB queries to reduce API response time by 20%.
- Built a **React and TypeScript-based** web application and a companion **Android** app that has a shared backend API layer, designing the backend to support multi-client access with consistent logic and scalable deployment on **AWS (EKS, EC2, S3, CloudFront)**.

Multi-LoRA(Low Rank Adaption) Integration Pipeline

- Developed a **multi-adaptor LoRA** blending system for diffusion models to preserve character identity across AI-generated images, reducing feature mixing by 58%.
- Implemented region-based attention control and adaptor blending on the **FLUX diffusion model** and validated improvements using **CLIP alignment** and **ArcFace** identity metrics across large-scale GPU experiments.

Research Papers

- **Trip Planner and Recommender using Flutter and Tensorflow** - "IEEE, 2022"
- **Conceptual Model Simulation to Detect and Report City Traffic Violations Using Distributed Intelligent Agents** - "IEEE, 2021"

Skills

Languages: Python, Java, C, C++, SQL, JavaScript, TypeScript, React.js, Angular, C#, Dart

Core: Distributed Systems, Multithreading, Memory Management, REST APIs, GraphQL

Backend: Node.js, Flask, Django, Express.js, Event-Driven Architecture, WebSockets

AI Systems: Generative AI(LLMs), RAG, LangChain, LangGraph, Vector Databases (Qdrant), Agentic AI, Diffusion Models

Cloud & DevOps: Google Cloud(GCP), AWS(EKS, EC2, S3, Lambda), Microsoft Azure, Docker, Terraform, CI/CD(Github Actions)

Data Engineering: PySpark, Hive, PostgreSQL, MongoDB, Redis, PL/SQL, MySQL